

Spatio-Temporal summation of sequential inputs and their propagation along
a dendritic branch in a Multi-Compartmental Model using MOOSE

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ABSTRACT

This project explores how sequential input via current injection and synaptic input along a model cylindrical dendrite affects the membrane potential at the soma, using a biophysical neuron model based on Hodgkin-Huxley dynamics. The goal was to develop an intuitive and computational understanding of dendritic signal integration and the electrotonic structure of neurons. A simplified cylindrical morphology was constructed and divided into 10 compartments to represent the dendrite, with the last compartment representing the soma. Active and passive membrane properties were implemented using the Hodgkin-Huxley formalism within the MOOSE simulator. Sequential current injections were applied at increasing distances from the soma in both proximal-to-distal and distal-to-proximal directions to examine how input location influences the amplitude and timing of the somatic response. The simulations demonstrated a clear distance-dependent attenuation of membrane potential, consistent with cable theory and known biophysical principles, along with spatio-temporal summation of signals. The model with voltage-gated channels demonstrated propagation of action potentials. This work deepened my understanding of neuronal excitability, signal propagation, and the role of dendritic morphology in shaping neuronal output. The model provides a foundation for more complex studies involving multiple channel types, non-uniform channel distributions, and realistic morphologies. Conversely, detailed morphologies can be simplified into idealized structures such as this, allowing for intuitive and computationally inexpensive simulations to be performed prior to developing more elaborate models.

INTRODUCTION

Neurons receive and process information through complex patterns of electrical activity, with dendrites playing a crucial role in determining how synaptic inputs are integrated and transmitted to the soma. The spatial and temporal distribution of synaptic inputs across a neuron's dendritic tree significantly shapes the resulting membrane potential at the soma, influencing whether an action potential is triggered. Understanding how these inputs are shaped by the biophysical properties of the neuron is critical for understanding how a neuronal circuit behaves and for gaining insight into neuronal computation.

Dendrites are not just passive conduits of current; their membrane contains a variety of ion channels, both passive and active, that can influence how signals propagate. In particular, the profile of the voltage signals as they travel from distal dendrites to the soma is governed by both passive cable properties and the distribution of active channels such as voltage-gated sodium, potassium and calcium channels.

In this project, the aim was to develop a compartmental model of a simplified cylindrical dendrite and soma, in order to investigate how sequential synaptic-like inputs at increasing distances from the soma affect the membrane potential at the soma. The model was implemented using the MOOSE (Multiscale Object-Oriented Simulation Environment) simulator and incorporates Hodgkin-Huxley dynamics to simulate realistic neuronal behavior.

By simulating current injections at various dendritic locations in both proximal-to-distal and distal-to-proximal directions, key principles of dendritic integration, such as signal attenuation, summation, and action potential propagation were observed. This simplified setup serves as a foundation for exploring more biologically detailed morphologies and channel distributions in future work.

OBJECTIVES:

The main goal is to understand how signals propagate along a dendritic branch and how these signals undergo spatio-temporal summation at the soma.

In this project, three separate simulations are to be carried out as specific work plan:

1. Only passive compartments devoid of any voltage gated channels are to be considered. Current injections are to be set up at each compartment and activated in a sequential manner to see propagation of the signals and their summation at the soma. This simulation is supposed to bring to light the cable properties of neurons.
2. In this simulation, the setup is to be kept the same as the first but instead of current injections, a synaptic input is to be set up to simulate more realistic current injections (EPSC) into the compartment. This is done to understand propagation of more realistic EPSP's (depolarizations due to influx via synaptic channels)
3. Finally, in the third simulation, voltage gated channels are to be added to study the propagation of action potentials.

METHODS AND RESULTS

Since MOOSE is object oriented, it provides multiple classes necessary for neuronal simulations. Objects are created from these classes and can be connected up in the desired manner and then the simulation can be carried out.

Instances of the following classes were used:

Compartment : 'Compartment'

Pulse Generator : 'PulseGen'

Spike Generator : 'SpikeGen'

Voltage gated Channels : 'HHChannel'

Table for recording : 'Table'

CONSTRUCTING THE MODEL:

A total of 10 compartments were created and connected back to back and the last compartment was treated as the soma (Scheme 1). The compartments were cylindrical in shape and the membrane resistance and capacitance were set based on the surface area of the compartments.

The properties of the compartments are as follows:

(All the values are assigned in terms of SI units)

`erest=-70e-3     ## resting membrane potential`

`radius=5e-6     ## we can set 2*radius in the diameter field of each compartment`

`length=20e-6`

`Cm_spec=1e-2    ## Membrane capacitance = Specific Capacitance * sarea`

`Gm_spec=3       ## Membrane Conductance = Specific Conductance * sarea`

`sarea=2*np.pi*radius*length   # Surface Area`



Scheme 1: The base structure of the model that was used in all of the simulations.

SIMULATION 1:

Understanding the propagation of signals (due to current injection) in passive compartments and their summation in the soma.

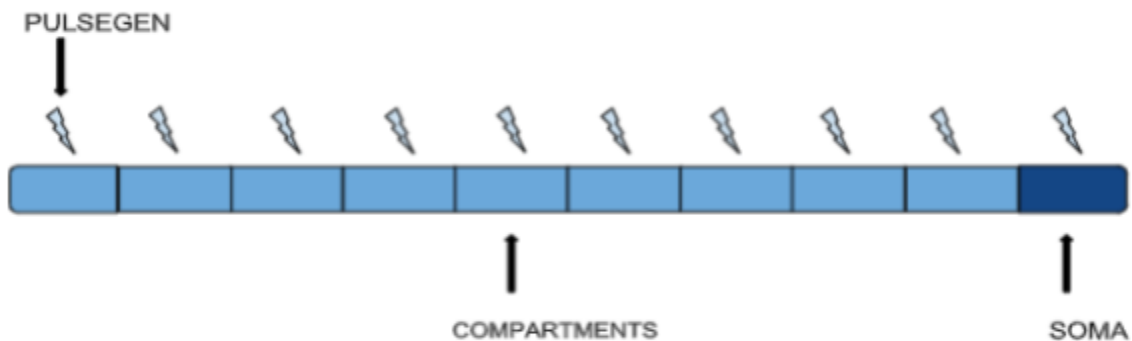
10 Pulse Generators were connected to each of their corresponding compartments. The Pulse generators were used to inject a current into each of the compartments in a sequential manner (Scheme 2).

The parameters of the Pulse Generator used are as follows:

`pulsegen.level[0]=100e-12` `##` Current injected was 100 pA

`pulsegen.width[0]=20e-3` `##` The current was injected for a duration of 20 ms

To achieve sequential activation of Pulse Generators, a ‘for’ loop was used to assign progressively increasing values to the ‘firstDelay’ field of each Pulse Generator. This ensured that each generator was triggered immediately after the preceding one.



Scheme 2: A schematic representation of the system used for simulation 1

The runtime of the simulation was 250 ms and the first Pulse Generator activated after a 10 ms delay in the beginning. Then each Pulse Generator activated immediately after the previous one stopped injecting current and each of them delivered a sustained output value (which in this case serves as the current output) for 20 ms.

The plots of the current output and the effect of those current injections on the somatic membrane potential is shown in Figure 1.

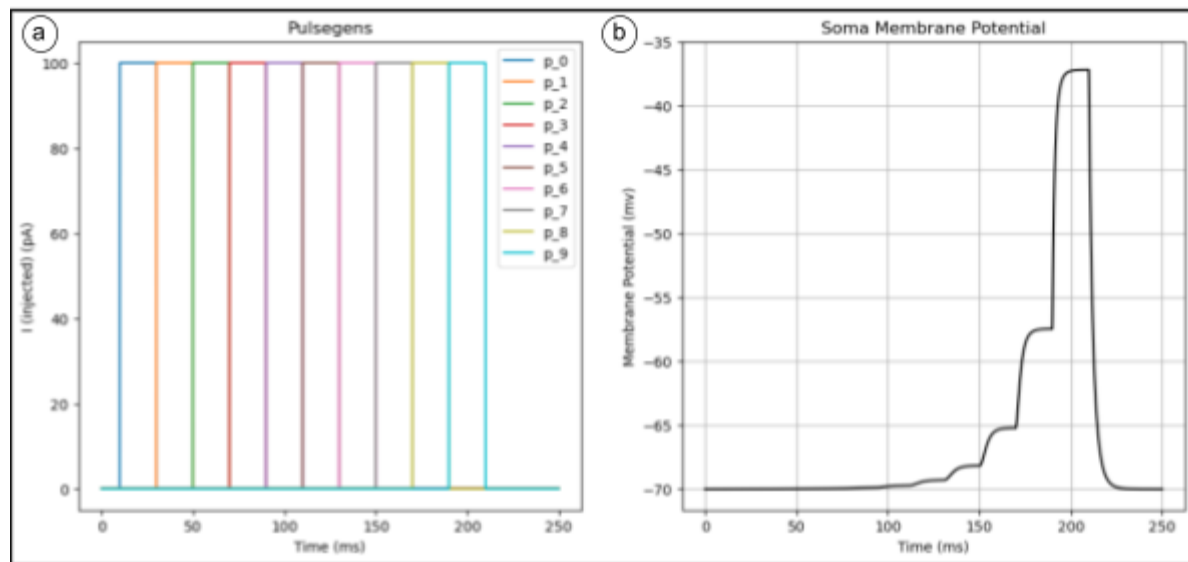


Figure 1:

a Plots of the current output by the Pulse Generators. All of the current outputs have an amplitude of 100 pA and duration of 20ms. The blue plot represents the Pulse Generator connected to the first compartment and is labelled as p_0 and the indigo plot labelled as p_9 is the Pulse Generator connected to the soma.

b Plot of the membrane potential at the soma. From this plot we can see the summation of signals flowing into the soma from the previous compartments due to sequential current injection. The plots follow a profile similar to that of the charging of a capacitor, illustrating that the cell does indeed behave like a capacitor.

The same simulation was carried out but this time in a proximal to distal manner. The plots of the Pulse Generators and membrane potential at the soma is shown below in Figure 2.

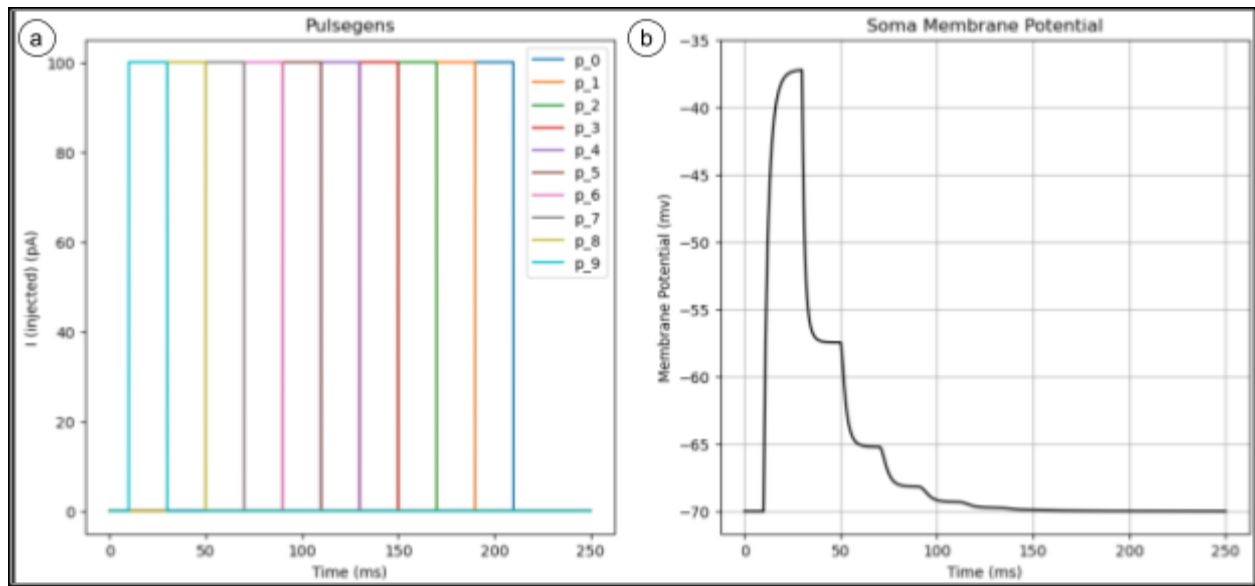


Figure 2:

a Plots of the current output by the Pulse Generators. They have the same parameters but in this case, the activation order was reversed. p_9 fired first and p_0 fired last.

b The plot of the membrane potential at the soma. The plot is characteristic of the discharging of a capacitor.

In the plot of the membrane potential, the potential drops exponentially but flattens out at the times the signals from the neighbouring compartments flow in. The flattening out is due to the overlap of the two profiles, one being an **exponential decay** (discharging of the capacitor) and the other an **exponential rise** (charging of the capacitor due to signal flow).

It might not be evident from the picture, but there is a difference in the peak amplitude of the membrane potential between the two cases.

When the signal input is moving from distal to proximal compartments, the peak potential is **-37.2142** and when moving in the proximal to distal direction, the peak potential is **-37.2426**. Although subtle, the peak potential is greater when moving from distal to proximal end, illustrating the property of signal summation as they are summing up to give a higher membrane potential.

In the second part of the same simulation, only the Pulse Generator connected to the 5th compartment (p_4) was activated for the same duration of 20 ms and the membrane potential change for the 5th-10th compartment were recorded and plotted to show the propagation and attenuation of the signal (Figure 3).

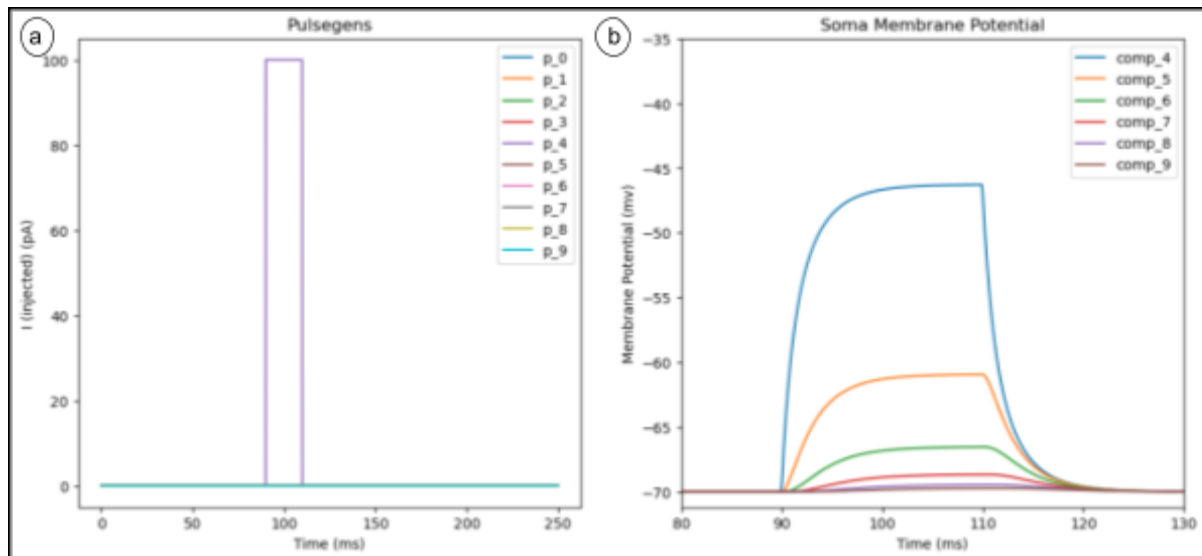


Figure 3:

a Plot showing the activation of the Pulse Generator connected to the 5th Compartment

b In this figure, we can see the membrane potential variation of the 5th compartment (labelled comp_4) and the compartments after that till the soma (labelled as comp_9).

Since the current is being injected into the 5th compartment, the maximum change in membrane potential is seen for the 5th compartment itself and then the membrane potential changes for the subsequent compartments are seen as well but with lower amplitudes. This is due to the attenuation of the signal as it flows from one compartment into the next. The attenuation of signal amplitude is primarily due to axial resistance, which originates from the cytoplasmic resistivity. This resistance causes voltage to drop as the signal propagates along the dendrite.

It can also be noticed that the rise of the curve is more and more delayed as the signal propagates further highlighting the time taken for the signal to propagate.

SIMULATION 2:

Understanding the propagation of signals (due to synaptic input) in passive compartments and their summation in the soma.

For this simulation, the 10 Pulse Generators were connected to their corresponding Spike Generators which in turn were connected to the synapses of their respective compartments.

A Spike Generator simulates a spike event in a presynaptic neuron.

In a biological neuron, once the action potential reaches the terminal end of the neuron, it causes calcium ion influx due to which neurotransmitters are released into the synaptic cleft. These neurotransmitters bind to the receptors and open the synaptic channels through which the specific ions can flow through to induce an EPSP in the postsynaptic terminal. In MOOSE, to simulate the effect of the neurotransmitters as a result of a presynaptic spike event, a **SpikeGen** object is created. If the membrane potential at the presynaptic end is high enough and crosses the threshold of the SpikeGen, then the SpikeGen is activated and sends a signal to the synapse which is handled by an object (synh) of the class **SimpleSynHandler**(this class sets the properties of the synapse) to open the synaptic channel which is represented by an object (synchan) instantiated from the class **SynChan** and allow the flow of ions to essentially depolarize the postsynaptic compartment (Scheme 3).

For this experiment, to simulate the presynaptic terminal, the Pulse Generators have been used. If the amplitude of the output value being sent out is high enough, then it will activate the SpikeGen which will then send a signal to open the synaptic channels (Scheme 4).

(Note: It might seem counterintuitive to use a PulseGen to mimic presynaptic membrane potential but in a computer simulation, these just serve as numbers and can be used as substitutes even though they do not make physical sense.)

The parameters used in the simulation are as follows:

```
pulsegen.level[0]=1e-10  ## Amplitude of current (proxy for presynaptic membrane potential)
pulsegen.width[0]=20e-3  ## Duration of activation of the Pulse Generator
```

```

proto_spikegen.threshold=5e-11 ## The threshold of the Spike Generator.

synh.numSynapses=1  ## Number of synapses per compartment

synchan.Gbar=1e-6    ## Maximum conductance of the synaptic channel

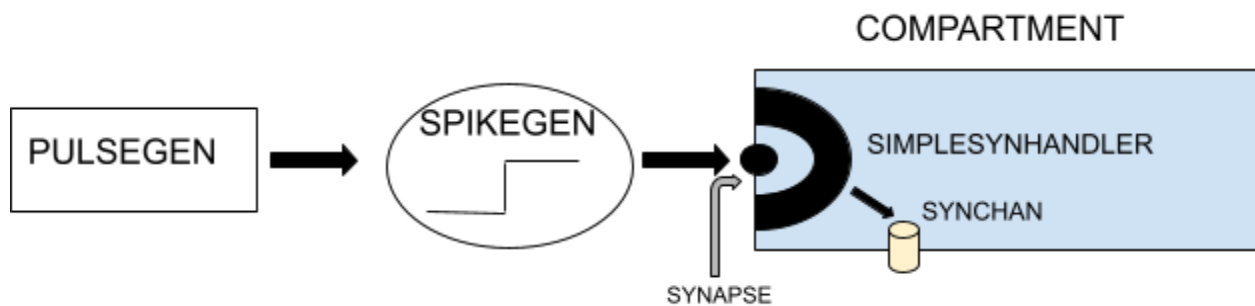
synchan.Ek=0.0       ## Reversal potential of the channel

synchan.tau1=1e-3    ## tau1 and tau2 are the rise and decay time constants for the synaptic
synchan.tau2=1e-3    channel. tau1 = tau2 for an  $\alpha$  synapse

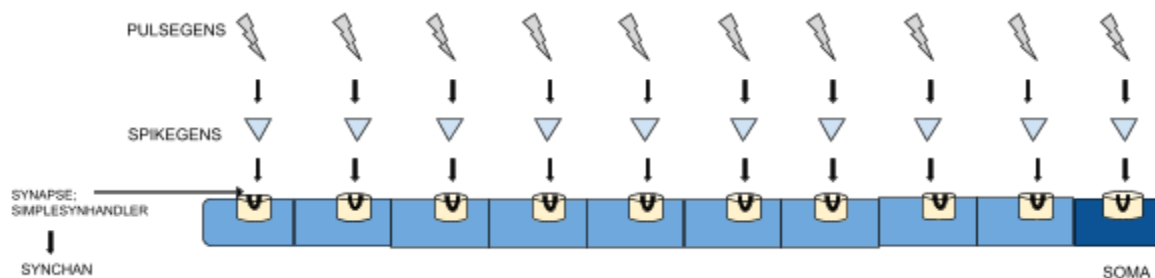
synh.synapse[0].weight=1  ## Degree of influence of presynaptic terminal on postsynaptic end
synh.synapse[0].delay=1e-3 ## Delay between the time Spike is sent and opening of channels.

```

The delay is necessary to mimic the diffusion time needed for the neurotransmitter to diffuse across the synaptic cleft.



Scheme 3: A visual representation of the flow of information



Scheme 4: A schematic representation of the model used for the simulation

There were two parts to this simulation. In part 1, the delay of the Pulse Generator was kept at 15 ms to let the membrane potential always come back to rest. This was done to examine the peak potential when the Pulse Generators were activated in a distal-to-proximal manner and vice versa.

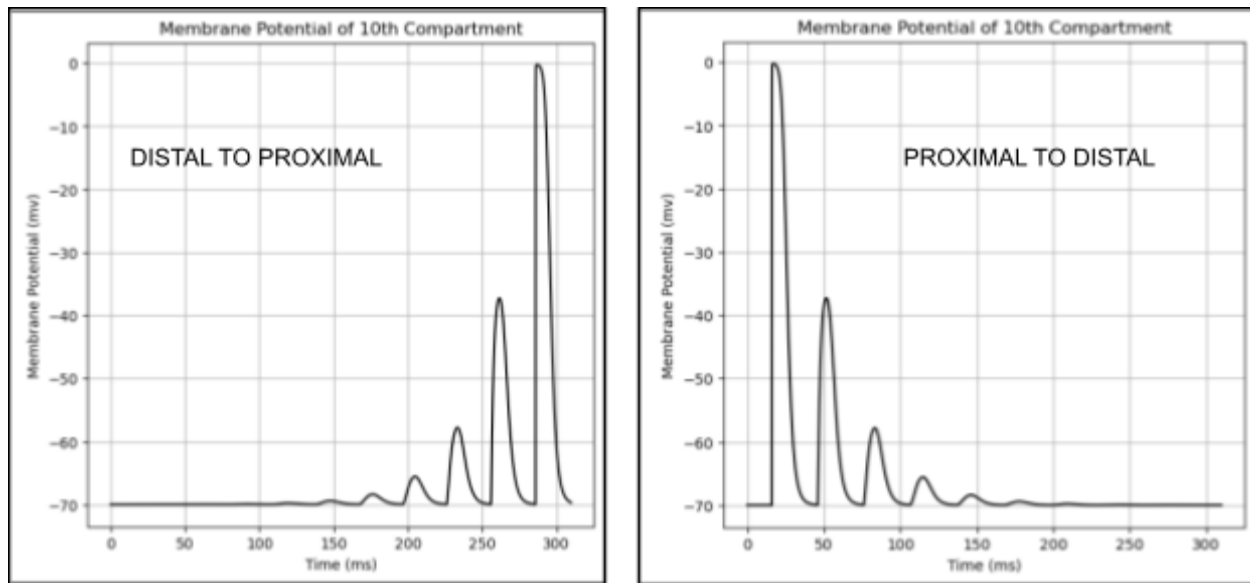


Figure 4: The delay of the Pulse Generators were set to 15 ms to allow complete repolarization of the cell

a Plot of Membrane Potential when Pulse Generators are activated in a distal to proximal manner

b Plot of Membrane Potential when Pulse Generators are activated in a proximal to distal manner

A clear distinction between this and SIMULATION 1 can be seen in Figure 4. It is evident that it is not a sustained current input from the way the potential goes up and then decays down again indicating that the synaptic channels activate rapidly and then the potential is decreased again by the action of the leak channels. Similar attenuation of signals flowing in from the neighbouring compartments can be seen in both the plots.

The peak potential in both cases came out to be -0.238 mv confirming that the direction of sequential input does not matter as long as the cell repolarizes to its resting membrane potential.

In the second part of the simulation, the delay of the PulseGen was decreased to 5ms to show the summation effect of the signals.

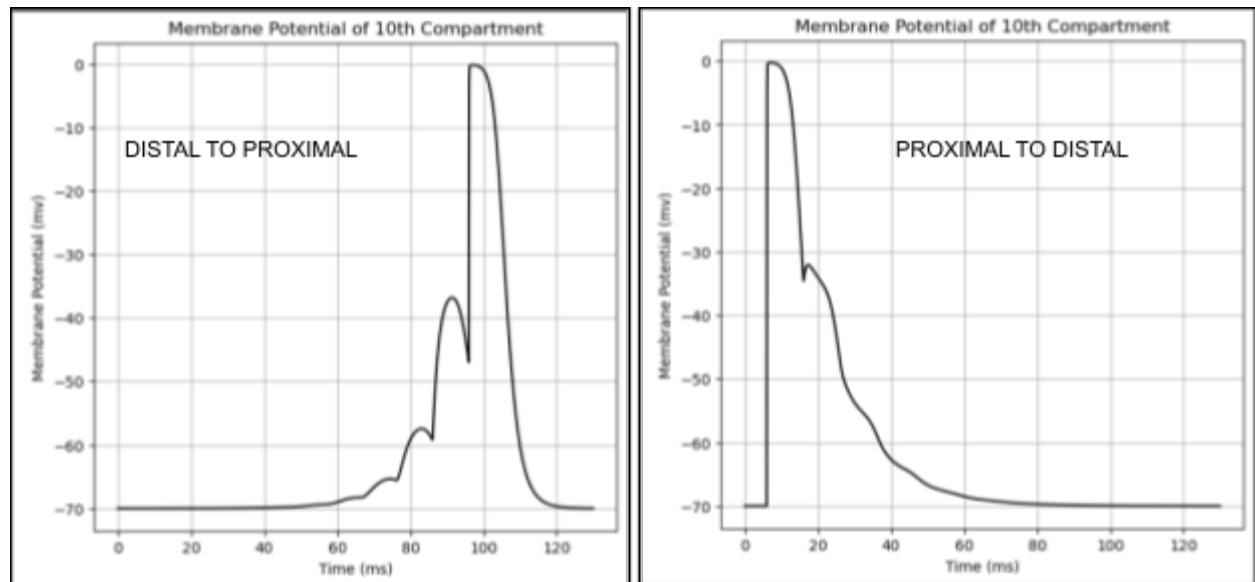


Figure 5: The delay of the Pulse Generators were set to 5 ms to allow complete repolarization of the cell

a Plot of Membrane Potential when Pulse Generators are activated in a distal to proximal manner

b Plot of Membrane Potential when Pulse Generators are activated in a proximal to distal manner

In this case (as depicted in Figure 5), a difference in peak membrane potential was observed. When the signal input moved from the distal to proximal end(Figure 5a), there was a summing up effect which increased the membrane potential to -0.201 mV which is higher than the peak potential in Figure 5b recorded at -0.238 mV.

SIMULATION 3:

The model was further modified to include voltage-gated ion channels this time. The passive components were kept the same along with the synapses, Pulse and Spike Generators.

The voltage-gated ion channels are modelled using Hodgkin-Huxley type equations.

In this type of model, an ion channel used in the compartmental model is modelled based on voltage clamp recordings for a particular type of ion channel. Based on the voltage clamp

recordings, curve fitting is done to match the conductance curve of that type of ion channel at different voltages. The variables that are fit to the curve are α and β which are essentially rate constants for opening and closing respectively, of a gate of the voltage-gated ion channels. The potassium channel has an **activation(n)** gate while the **sodium** voltage gated channel has two types of gate: **activation(m)** gate and **inactivation(h)** gate.

Here, **n,m and h** represent the fraction of the gate open.

The rate of opening of a voltage gated ion channel can be expressed in the form of a first order differential equation in the form of:

$$dx/dt = \alpha(1 - x) - \beta x ; \text{ where } x \in \{n,m,h\}$$

Upon solving the differential equation, we can express x in terms of x_{∞} corresponding to the value of x at steady state ($t=\infty$), x_0 corresponding to the value of x at initial time and τ which is the time constant of the exponential.

$$x(t) = x_{\infty} + [x_0 - x_{\infty}] e^{-t/\tau}$$

where

$$x_{\infty} = \alpha / (\alpha + \beta)$$

$$\tau = 1 / (\alpha + \beta)$$

In MOOSE, the expression of α and β are manually assigned and then their values are calculated at different voltage steps in a particular range and assigned to tables specific to a particular gate. From these tables, x_{∞} and τ are calculated during the time of the simulation and in accordance to these values, the conductance at a given particular voltage are calculated using the formulae $g = g_{\max} n^4$ for K voltage gated channel and $g = g_{\max} m^3 h$ for Na voltage gated channels.

The parameters used in the simulation are:

`erest = -70e-3                ## resting potential of compartment`

`ENa = 115e-3 + erest        ## Sodium reversal potential`

`EK = -12e-3 + erest        ## Potassium reversal potential`

`nachan.Gbar = 120 * 10 * sarea ## 120 mS/cm^2 in HHM. In SI, we multiply by 10`

`kchan.Gbar = 36 * 10 * sarea # 36 mS/cm^2 in HHM.`

Voltage steps are setup in the following manner:

```
vmin, vmax, vdivs=-100e-3, 100e-3, 2000
```

```
V = np.linspace(vmin,vmax,vdivs) ## an array containing the voltage steps
```

The different expressions for alpha and beta of different gates:

```
m_gate_alphaExpr = 1e3*0.1 * ( 25 - 1e3 * (v - (erest)))/(np.exp((25 - 1e3 * (v - (erest)))/10) - 1)
```

```
m_gate_betaExpr = 1e3 * 4 * np.exp(- 1e3 * (v - (erest))/ 18)
```

```
h_gate_alphaExpr = 1e3 * 0.07 * np.exp(- 1e3 * (v - (erest))/ 20)
```

```
h_gate_betaExpr = 1e3 / (np.exp((30 - 1e3 * (v - (erest))) / 10) + 1)
```

```
n_gate_alphaExpr = 1e3 * 0.01 * (10 - 1e3*(v-(erest))) / (np.exp((10 - 1e3 * (v - (erest)))/10) - 1)
```

```
n_gate_betaExpr = 1e3 * 0.125 * np.exp(-1e3 * (v - (erest)) / 80)
```

The multiplication by 1e3 is to convert volts to mv during calculations and the potentials have been scaled down by 'erest' to account for the modern electrophysiological conventions as compared to the conventions used by Hodgkin and Huxley in their paper.

(Note: The equations for the alpha and beta expressions are the original equations published by Hodgkin and Huxley in their 1952 paper on current conduction in the giant axon of Loligo.)

The first Pulse Generator fired after 10 ms and sustained a current output for 20 ms. The next Pulse Generator fired immediately after the first one stopped giving a current output. The membrane potential variation was measured at the soma.

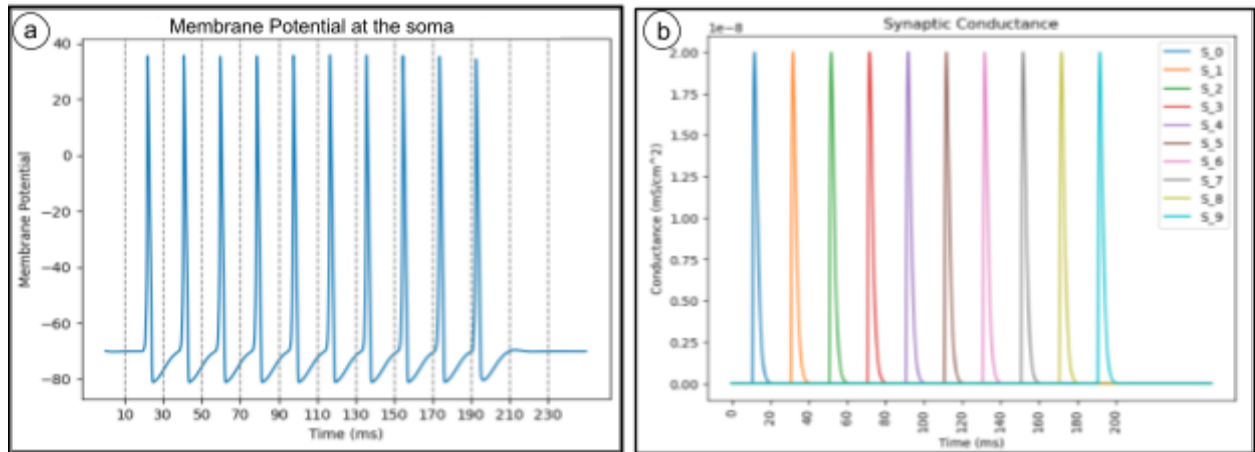


Figure 6:

a Variation of membrane potential at the soma due to sequential spike stimulus in a distal to proximal manner. The dotted lines indicate the times at which the Spikes were sent out to the synapse. The first line corresponds to the spike received by the synapse at the first compartment, the second line represents the time at which the spike was received by the second compartment and so on till the 10th compartment which is the soma.

b Conductance profile of the synaptic channels showing that their conductances increase at the spike times.

In Figure 6, we see action potentials flowing into the soma. The earliest action potential plotted is the action potential flowing in from the first compartment. Since the compartments have voltage-gated channels, they are capable of generating action potentials which flow into the neighbouring compartments and are further maintained by the voltage-gated channels in the neighbouring compartments. In the figure, we can see the relative shift from the dotted lines decrease as the time increases - a testament to the time needed for the flow of signals. The earliest action potential needs the longest time to flow from the 1st compartment to the 10th. The action potential generated at the 2nd compartment requires less time and eventually we can see that the action potential is generated immediately when the stimulus is at the 10th compartment. The variation of the membrane potentials at all the compartments were measured for the activation of only the first Pulse Generator and a time lag was seen in between action potentials in the following compartments (figure not shown). In this simulation, the signals did not attenuate as a result of having the voltage gated channels which produced action potentials of the same amplitude in each of the compartments.

Bi-directional propagation of signals were also accounted for.

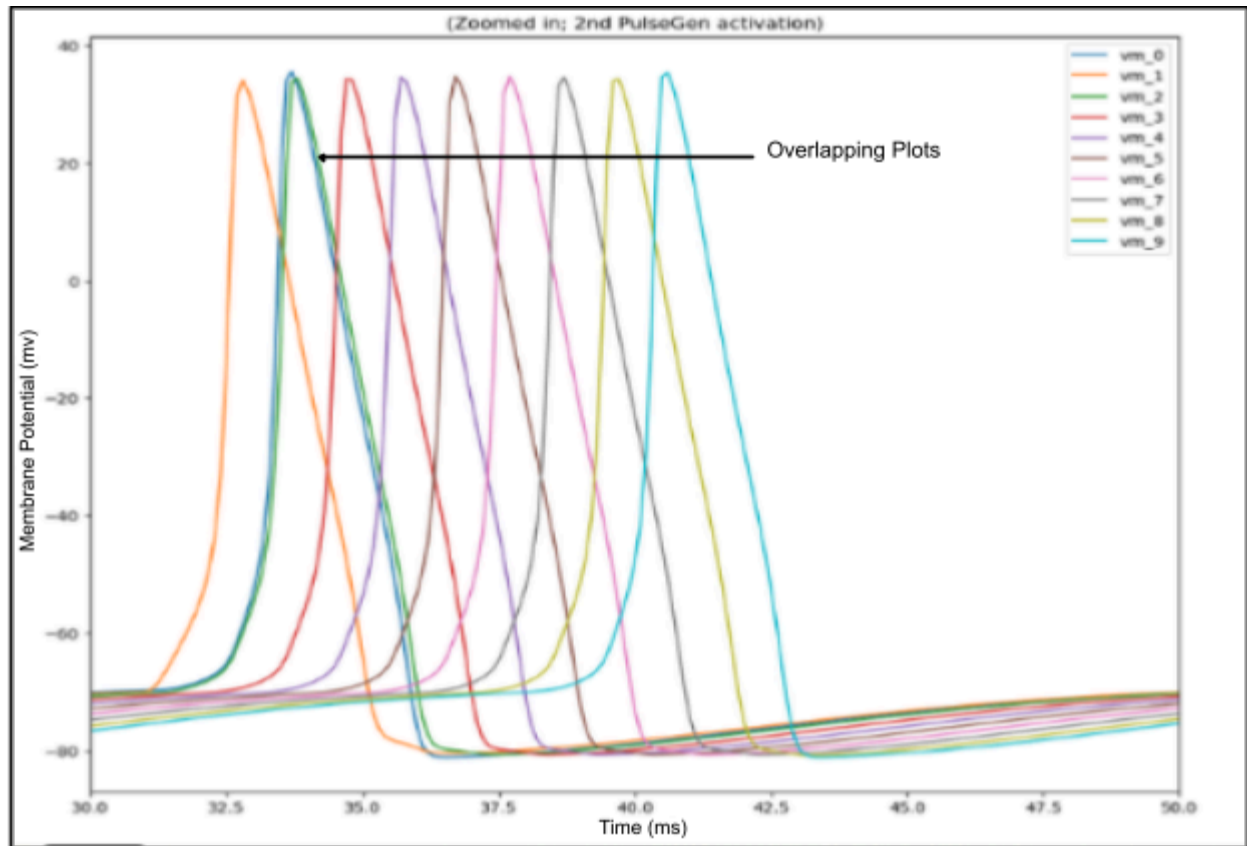


Figure 7: Plot of the membrane potential of all the compartments when the Pulse Generator connected to the Spike Generator connected to the 2nd compartment was activated. A demonstration of bi-directional flow of signal.

In Figure 7, it is seen that the membrane potential of the 1st(labelled vm_0) and 3rd(labelled vm_3) compartment overlap with each other. The first increase in potential is seen in the 2nd compartment (vm_1) as that is where the spike is initiated. Then, it proceeds to flow in both directions leading to generation of action potentials in compartments 1 and 3 simultaneously. This is an illustration of bi-directional propagation of action potentials.

DISCUSSION:

The simulations paint a picture of the propagation of signals and their spatio-temporal summation. This is in accordance with the literature published by Wilfrid Rall showcasing the electrical properties of dendrites. The results of such simulations can be used to analyse the effect of synaptic input at the soma and overall generation of action potentials. Hebbian plasticity dictates that ‘neurons that fire together wire together’, so varying synaptic strengths can have varying effects on the somatic membrane potential. Additionally, when we consider the complex morphology of a neuron, summation of signals at certain sites having synaptic inputs could potentially alter synaptic plasticity by activating the Ca^{2+} driven mechanism for expressing more AMPA receptors. Prolonged sub-threshold EPSP’s at synaptic sites affects certain ion channels dynamics in certain ways which allow back-propagating action potentials to invade those sites preferentially leading to increase in synaptic strength through certain mechanisms - a simplified version of such events as this can be modelled using the same sort of model with modifications in channel types and dynamics. Timings of external input play a crucial role in plasticity as seen in STDP (Spike Time Dependent Plasticity). We also observed bi-directional propagation of signals; in dendrites this signal has to be attenuated very rapidly as it moves away from the soma. For this purpose, A type K^{+} channels are present in the CA1 pyramidal neurons of the hippocampus and these channel dynamics can be incorporated into the model to study the effect of such channels.

The concept of signal summation in biological neurons has inspired developments beyond biology, most notably in the field of Artificial Neural Networks (ANNs). In ANNs, computational units called *neurons* mimic this behavior by aggregating input signals through a weighted linear combination, followed by the application of an activation function to produce an output. These artificial neurons form the basis for training models used in tasks such as object detection—an application that, interestingly, has found its way back into biological research through areas like image-based cell classification and behavioral tracking.

CONCLUSIONS:

To conclude, the model used for the simulations successfully displays the electrical properties of the neuron along with Hodgkin-Huxley channel dynamics which describes the flow of ions through the ion-specific voltage gated channels.

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7. Moose-examples : A Github having example code for different types of simulations on MOOSE (<https://github.com/subhacom/moose-examples>)

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Finally, I acknowledge the broader faculty at CHINTA for fostering an environment of inquiry and learning.

SELF-REFLECTION:

The initial weeks of my internship focused on training, where I learned the foundational concepts of neuroscience - how neurons behave, the types of ion channels, their functions, and how these vary across different regions of the nervous system. This was followed by an introduction to the mathematical modeling of channel dynamics based on the Hodgkin-Huxley framework and its implementation using the MOOSE simulation environment.

This journey was far from easy, and I made several mistakes along the way. However, these challenges taught me the value of critical analysis and the importance of asking questions even if they seem obvious. I discovered that learning is a continuous process; one question often leads to many more. What stood out most was the importance of being curious and daring, exploring unfamiliar domains and realizing how knowledge from one field can creatively inform another. This mindset to explore, ask questions and critically assess information has been my most valuable takeaway from the experience.

CERTIFICATION:

This is to certify that the project titled “**Spatio-Temporal Summation of Sequential Inputs and Their Propagation Along a Dendritic Branch in a Multi-Compartmental Model using MOOSE**” submitted by **Pramesh Chakraborty** (Reg. No: A01-1112-0664-24) to the Department of Chemistry, St. Xavier’s College (Autonomous), Kolkata, is a record of the work carried out during the Summer Internship (May–July 2025), under my supervision.

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Date: _____